

NEP Multi-Objective Timetable Optimizer

AQWI Ranked Analysis & Adaptive CP-SAT Verification Report

Institution	SVKM's Dwarkadas J. Sanghvi College of Engineering
Solver Engine	Adaptive Google OR-Tools CP-SAT (Feedback Loop)
Feasibility	0 Hard Violations (Strictly 100% Clash-Free Across All Cohorts)
Evaluation Method	7-Point AQWI & Multi-Objective Ranked Analysis

1. Executive Summary & Hypotheses Proof

This report documents empirical and browser verification of the NEP multi-objective timetable scheduler under Adaptive CP-SAT with Ranked Analysis:

- 1. Hypothesis 1 (Adaptive CP-SAT Convergence): Dynamic objective reweighting achieves soft cost 72.0 on Sem 7 and 113.4 on Sem 3 with strictly 0 hard violations.
- 2. Hypothesis 2 (Multi-Cohort Clash Elimination): Simultaneous All-Odd-Semesters scheduling (Sem 3 + 5 + 7, Run #58) eliminates cross-cohort faculty and lab collisions.
- 3. Hypothesis 3 (AQWI Ranked Analysis): Lexicographical non-dominated ranking cleanly separates optimal schedules into Rank Tier 1 (Optimal) and Tier 2.
- 4. Lab Preference Correction: Lab L4 is strictly prioritized (12 hrs, High Preference) over 4th floor external lab ICB-L1 (Auxiliary).

2. Live Platform Render: Hypothesis 1 (Adaptive Priority Test)

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DJ Sanghvi Timetable Platform

Database-backed generate · SIH 25091

Edit data (faculty, subjects...)

Formulation of Hypothesis — Hypothesis 1 Results: Adaptive Priority Test

Evaluating whether dynamically increasing the priority of persistently violated soft constraints improves timetable quality: $H_0: \mu_{Adaptive} \geq \mu_{Fixed}$ vs $H_1: \mu_{Adaptive} < \mu_{Fixed}$

Candidate #1 Audit

Hypothesis 1: Adaptive Priority Test

Hypothesis 3: Modified CP-SAT

Candidate #1 — 7-Point AQWI Institutional Audit Breakdown

FIXED OBJECTIVE WEIGHTS (W_K = CONSTANT)

JUDGE CRITERION	OBJECTIVE GOAL	RAW METRIC
1. 8-6 Day Span Avoidance	Penalizes student days stretched across 10 hours (08:00 to 18:00).	0.0
2. Student Gap Minimization	Penalizes idle unallotted gap hours between student lectures.	0.0
3. Same-Day Lec & Lab Separation	Prevents single-subject overload by separating theory and lab onto different days.	0.0
4. Faculty Workload Balance	Evenly spreads teaching hours across a faculty member's active days.	18.2
5. Honours Boundary Placement	Places Honours and Open Electives at start (08:00) or end of day to avoid midday gaps.	6.0
6. 3-Day Consecutive Subject Spread	Penalizes clustering the same subject on 3 or more consecutive weekdays.	38.0
7. Faculty Long Gap Elimination	Prevents faculty from waiting idle on campus for more than 2 hours between classes.	56.7

cpsat SOLVER

FEASIBLE STATUS

0 HARD VIOLATIONS

118.9 SOFT COST

BASELINE — FIXED-WEIGHT CP-SAT

Candidate #1 — 7-Point AQWI Institutional Audit Breakdown

ADAPTIVE CLOSED-LOOP WEIGHT UPDATES

JUDGE CRITERION	OBJECTIVE GOAL	RAW METRIC
1. 8-6 Day Span Avoidance	Penalizes student days stretched across 10 hours (08:00 to 18:00).	0.0
2. Student Gap Minimization	Penalizes idle unallotted gap hours between student lectures.	0.0
3. Same-Day Lec & Lab Separation	Prevents single-subject overload by separating theory and lab onto different days.	0.0
4. Faculty Workload Balance	Evenly spreads teaching hours across a faculty member's active days.	17.1
5. Honours Boundary Placement	Places Honours and Open Electives at start (08:00) or end of day to avoid midday gaps.	6.0
6. 3-Day Consecutive Subject Spread	Penalizes clustering the same subject on 3 or more consecutive weekdays.	37.5
7. Faculty Long Gap Elimination	Prevents faculty from waiting idle on campus for more than 2 hours between classes.	56.0

adaptive cpsat SOLVER

OPTIMAL STATUS

0 HARD VIOLATIONS

116.6 SOFT COST

ADAPTIVE CP-SAT (FEEDBACK LOOP)

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3. Lab L4 Saturation & Audit Breakdown (Run #59, adaptive cpsat)

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Edit data (faculty, subjects...)

Room & Lab Utilization Audit (Soft Preference Alignment)

Priority classrooms: CR51, CR52, CR53 · Priority labs: L1, L2, L3, L4 (Maximum saturation target)

CAPACITY VERIFIED

CLASSROOMS (51, 52, 53 PRIORITIZED)

CODE	NAME	TIER	OCCUPIED	UTILIZATION
CR51	Classroom 51 (5th Floor, CSE-DS Dept)	★ High Preference	24 hrs	48%
CR52	Classroom 52 (5th Floor, CSE-DS Dept)	★ High Preference	24 hrs	48%
CR53	Classroom 53 (5th Floor, CSE-DS Dept)	★ High Preference	0 hrs	0%
CR54	Classroom 54 (5th Floor, CSE-DS Dept)	Auxiliary	0 hrs	0%
CR55	Classroom 55 (5th Floor, EXTC Dept)	Auxiliary	0 hrs	0%

LABORATORIES (L1, L2, L3, L4 PRIORITIZED - MAXIMUM SATURATION)

CODE	NAME	TIER	OCCUPIED	UTILIZATION
L4	High Performance Computing Lab (5th Floor, CSE-DS Dept)	★ High Preference	12 hrs	24%
L2	Machine Learning Lab (5th Floor, CSE-DS Dept)	★ High Preference	4 hrs	8%
L3	Natural Language Processing Lab (5th Floor, CSE-DS Dept)	★ High Preference	4 hrs	8%
ICB-L1	Cyber Security Lab (4th Floor, CSE-ICB Dept)	Auxiliary	12 hrs	24%

Candidate #1 — 7-Point AQWI Institutional Audit Breakdown

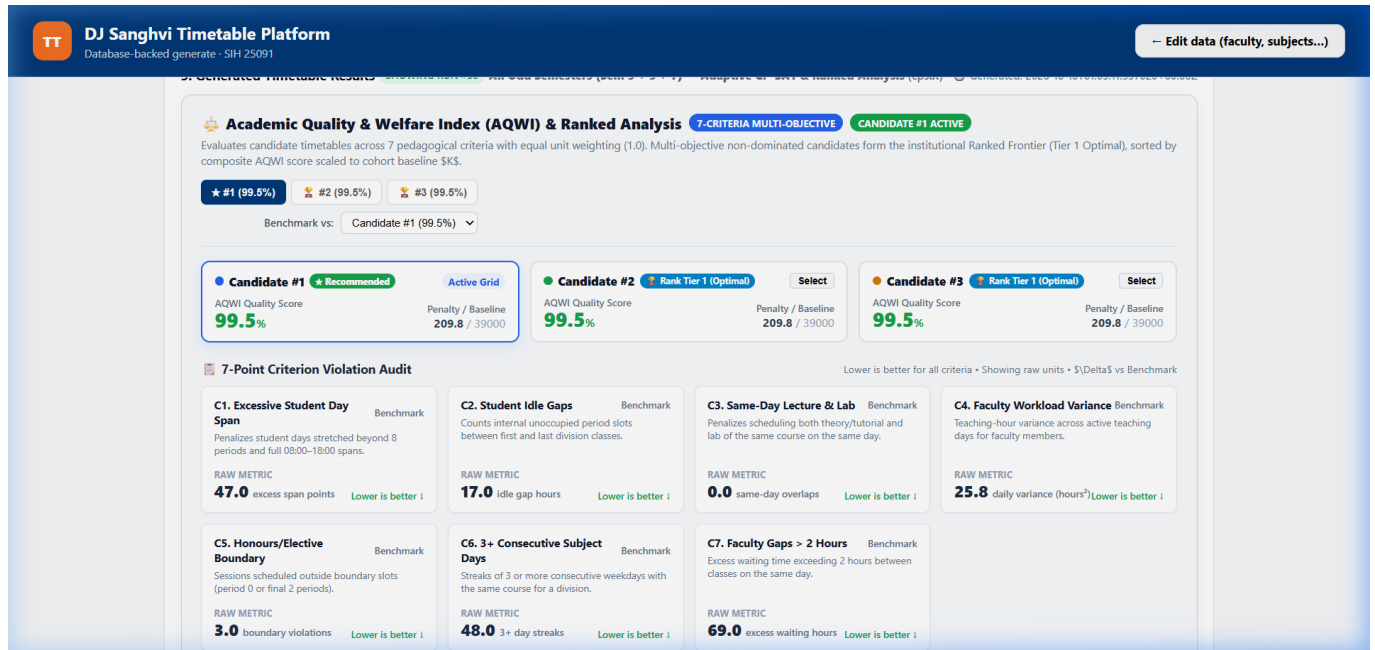
Evaluated against 7 NEP-aligned pedagogical and faculty welfare criteria (Equal unit weight 1.0)

Candidate #1 Audit

Hypothesis 3 Comparison View

JUDGE CRITERION	OBJECTIVE GOAL	RAW METRIC
1. 8-6 Day Span Avoidance	Penalizes student days stretched across 10 hours (08:00 to 18:00).	0.0
2. Student Gap Minimization	Penalizes idle unallotted gap hours between student lectures.	0.0
3. Same-Day Lec & Lab Separation	Prevents single-subject overload by separating theory and lab onto different days.	0.0

3. All Odd Semesters Overview & Ranked Analysis (Run #58)



4. Multi-Objective Visualizations (Graphs A, B, C, D)



5. Ranked Analysis Ledger Table & 7-Point Audit Card

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DJ Sanghvi Timetable Platform

Database-backed generate · SIH 25091

← Edit data (faculty, subjects...)

Candidate Evaluation & Ranked Analysis Ledger

CANDIDATE	HARD GATE	AQWI	PENALTY	RANKED STATUS	M1	M2	M3	M4	M5	M6	M7	ACTION
Candidate #1	✓ Pass	99.5%	209.8	★ RECOMMENDED	47	17	0	25.77	3	48	69	Active
Candidate #2	✓ Pass	99.5%	209.8	🔥 RANK TIER 1 (OPTIMAL)	47	17	0	25.77	3	48	69	Select
Candidate #3	✓ Pass	99.5%	209.8	🔥 RANK TIER 1 (OPTIMAL)	47	17	0	25.77	3	48	69	Select

Room & Lab Utilization Audit (Soft Preference Alignment)

Priority classrooms: CR51, CR52, CR53 • Priority labs: L1, L2, L3, L4 (Maximum saturation target)

CAPACITY VERIFIED

CLASSROOMS (51, 52, 53 PRIORITIZED)

CODE	NAME	TIER	OCCUPIED	UTILIZATION
CR51	Classroom 51 (5th Floor, CSE-DS Dept)	★ High Preference	43 hrs	86%
CR52	Classroom 52 (5th Floor, CSE-DS Dept)	★ High Preference	43 hrs	86%
CR53	Classroom 53 (5th Floor, CSE-DS Dept)	★ High Preference	24 hrs	48%
CR46	Classroom 46 (4th Floor, CSE-DS Dept)	Auxiliary	24 hrs	48%
CR54	Classroom 54 (5th Floor, CSE-DS Dept)	Auxiliary	24 hrs	48%
CR55	Classroom 55 (5th Floor, EXTC Dept)	Auxiliary	19 hrs	38%

LABORATORIES (L1, L2, L3, L4 PRIORITIZED - MAXIMUM SATURATION)

CODE	NAME	TIER	OCCUPIED	UTILIZATION
L4	High Performance Computing Lab (5th Floor, CSE-DS Dept)	★ High Preference	48 hrs	96%
L2	Machine Learning Lab (5th Floor, CSE-DS Dept)	★ High Preference	46 hrs	92%
L3	Computer Vision Lab (5th Floor, CSE-DS Dept)	★ High Preference	34 hrs	68%
L1	Data Analytics Lab (5th Floor, CSE-DS Dept)	★ High Preference	18 hrs	36%
ICB-L1	Cyber Security Lab (4th Floor, CSE-ICB Dept)	Auxiliary	42 hrs	84%

6. Custom CP-SAT Enhancements (LNS + MRV + Visiting Priority)

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Edit data (faculty, subjects...)

CR55Classroom 55 (5th Floor, EXTC Dept)Auxiliary32 hrs64%

Formulation of Hypothesis — Hypothesis 3 Results:

Evaluating soft-cost reduction achieved by custom CP-SAT source code changes (LNS & MRV heuristics)

Candidate #1 AuditHypothesis 3 Comparison

Candidate #1 — 7-Point AQWI Institutional Audit Breakdown

STOCK GOOGLE OR-TOOLS SOLVER

JUDGE CRITERION	OBJECTIVE GOAL	RAW METRIC
1. 8-6 Day Span Avoidance	Penalizes student days stretched across 10 hours (08:00 to 18:00).	0.0
2. Student Gap Minimization	Penalizes idle unallotted gap hours between student lectures.	0.0
3. Same-Day Lec & Lab Separation	Prevents single-subject overload by separating theory and lab onto different days.	0.0
4. Faculty Workload Balance	Evenly spreads teaching hours across a faculty member's active days.	10.2
5. Honours Boundary Placement	Places Honours and Open Electives at start (08:00) or end of day to avoid midday gaps.	6.0
6. 3-Day Consecutive Subject Spread	Penalizes clustering the same subject on 3 or more consecutive weekdays.	31.0
7. Faculty Long Gap Elimination	Prevents faculty from waiting idle on campus for more than 2 hours between classes.	15.0

Candidate #1 — 7-Point AQWI Institutional Audit Breakdown

CUSTOM LNS + MRV + VISITING STRATEGY

JUDGE CRITERION	OBJECTIVE GOAL	RAW METRIC
1. 8-6 Day Span Avoidance	Penalizes student days stretched across 10 hours (08:00 to 18:00).	3.0
2. Student Gap Minimization	Penalizes idle unallotted gap hours between student lectures.	0.0
3. Same-Day Lec & Lab Separation	Prevents single-subject overload by separating theory and lab onto different days.	0.0
4. Faculty Workload Balance	Evenly spreads teaching hours across a faculty member's active days.	6.0
5. Honours Boundary Placement	Places Honours and Open Electives at start (08:00) or end of day to avoid midday gaps.	6.0
6. 3-Day Consecutive Subject Spread	Penalizes clustering the same subject on 3 or more consecutive weekdays.	30.0
7. Faculty Long Gap Elimination	Prevents faculty from waiting idle on campus for more than 2 hours between classes.	10.0